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Campanian/Maastrichtian stage boundary in southern California: Resolution and implications for large-scale depositional patterns

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ABSTRACT

Detailed paleomagnetic study of Upper Cretaceous, predominantly deep-sea fan deposits from the Point Loma Formation (San Diego, California), situated within the allochthonous Peninsular Ranges terrane, has produced a magnetic polarity stratigraphy with one reversed magnetozone. Foraminiferal and megafossil (primarily molluscan) occurrences demonstrate that these strata are late Campanian to early Maastrichtian in age, and that the reversed magnetozone is almost certainly basal Maastrichtian Chron 32r. Recognition of this chron provides a direct correlation of the North Pacific province biostratigraphic zonations to the standard Cretaceous stages in Europe. The identification of Chron 32r indicates that for the available biostratigraphic zonations the Pacific slope megafossil Campanian/Maastrichtian stage boundary shows the best accord with the standard European stage boundary. Improved age relations for these strata suggest that their large-scale depositional patterns were strongly influenced by both local active-plate-margin tectonics and global eustatic factors during the Late Cretaceous.

INTRODUCTION

Exact dating and correlation of Cretaceous stratigraphic sequences in western North America is limited by the discontinuous nature of outcrops, repetitive lithofacies, and scarcity of diagnostic fossils. Efforts to develop a biostratigraphic zonation have provided a general understanding of the age of many of these strata, but, except for the Santonian/Campanian stage boundary (Ward et al., 1983; Fry et al., 1985), problems still exist in correlating these strata to the European stratotypes. Fry et al. (1985) and Morris et al. (1986) have shown that much of the Upper Cretaceous strata of southern and Baja California (Fig. 1A) have a stable detrital remanence (DRM), and they have identified intervals of reversed polarity in several locations. The development of a high-resolution magnetic polarity stratigraphy and the improved age resolution that this will provide for these Upper Cretaceous strata is necessary for the continued development of paleoecologic, sedimentologic, paleogeographic, and microplate-tectonic models for this region.

GEOLOGIC SETTING

In the San Diego area, shallowly dipping Upper Cretaceous marine strata of the Point Loma and Cabrillo Formations (Rosario Group) occur in nearly undeformed coastal sequences between La Jolla and Point Loma (Fig. 1, A and B). These strata are exposed in the limbs of an east-plunging syncline that has a fold axis located in Mission Bay. Progradational-retrogradational sediment cycles have been identified throughout these strata on the basis of vertical facies changes. A comparison of the La Jolla and Point Loma sequences shows a similar pattern of sediment retrogradation-progradation-retrogradation that provides a basis for direct lithostratigraphic correlation across the syncline (Nilsen and Abbott, 1981).

PALEOMAGNETIC PROCEDURES

Oriented block samples were collected from three sections of the Point Loma Formation: La Jolla Cove, Sunset Cliffs, and near the tip of Point Loma (Fig. 1B). All rock types were initially sampled at Sunset Cliffs, but mudstones and siltstones produced the most consistent results. Four specimen cubes (2 cm^3) were cut from each block using a circular table saw equipped with a dry masonry blade. Each specimen was thermally demagnetized at 50 °C intervals up to 400–600 °C, and the magnetic remanence after each step was measured on a Superconducting Technologies cryogenic magnetometer.

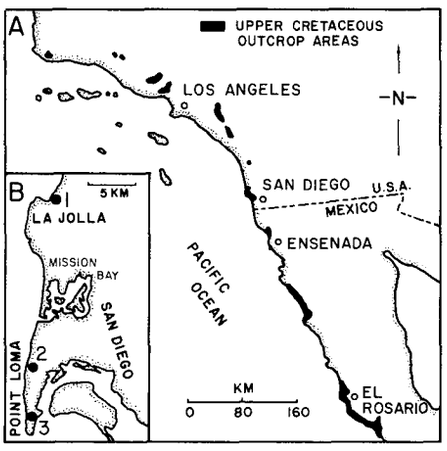


Figure 1. A: Upper Cretaceous outcrop areas in southern California and northern Baja California. Modified from Bottjer and Link (1984). B: San Diego location map showing sampled areas: 1, La Jolla Cove; 2, Sunset Cliffs; 3, Point Loma. Modified from Nilsen and Abbott (1981).

Zijderveld diagrams were used to determine the various characteristic remanences for each specimen (Fig. 2, A and B). In most cases, a normal-polarity secondary remanence, due either to a chemical remanence in authigenic goethite or a viscous remanence, was removed by 100–200 °C. This secondary component was particularly evident in reversed polarity samples (Fig. 2B) by a sharp drop in intensity up to 100–150 °C, and a rising intensity between 150 and 250 °C as the low-temperature overprint was removed. The primary remanence was typically demagnetized between 200 and 450 °C. The Curie temperature and X-ray diffraction studies indicate that the primary remanence is carried by a titanomagnetite. Occasionally, a small secondary remanence due to authigenic hematite remained after demagnetization to 500 °C.

Vector differencing of all specimens was used to obtain an average inclination, declination, and α_{95} for each site. The sites were then classified in the following manner: class 1 sites (N = 26) contained well-defined field vectors (α_{95} <20°) with good polarity information, class 2 sites (N = 27) contained less well-defined field vectors with good polarity, and class 3 sites (N = 5) contained an ambiguous record of mixed polarity and were discarded from further study.

Confidence that the final paleomagnetic results represent the original paleomagnetic field at the time of deposition is based on three tests. (1) Paleomagnetic results pass a reversal test (Fig. 2C) because the normal and reversed sites are antipodal. (2) The cleaned class 1 and 2 sites pass a fold test. The La Jolla Cove section lies north and the Sunset Cliffs and Point Loma sections lie south of the Mission Bay fold axis (Fig. 1B). The averaged inclination and declination across the syncline agree much better after correcting for the tectonic tilt associated with the

syncline (Fig. 2D), indicating that the paleomagnetic results record a prefolding magnetization. (3) A clear, replicable paleomagnetic polarity stratigraphy has been recovered from the three sampled intervals.

MAGNETOSTRATIGRAPHY

An interval of reversed polarity has been recognized in each of the three sampled sections (Fig. 1B). The reversed zone is best defined at La Jolla Cove where both the upper and lower boundaries are present. Seventeen sites were collected over 200 m of section in the slope and basin-plain and outer-fan lobe facies (Figs. 3 and 4). Five sites define a 55-m-thick reversed zone with the lower boundary approximately 60 m above the contact with the shallow-marine sandstone (Figs. 3 and 4).

The thickness of the reversed magnetozone on Point Loma cannot be measured directly because its upper and lower boundaries have not both been identified at any one location. The uppermost nine sites at Sunset Cliffs are reversed in polarity over 30 m of section, and the lower boundary of the reversed magnetozone is clearly defined 10 m above the base of the section. Farther south, near the tip of Point Loma, the upper boundary of the reversed magnetozone is well exposed in the hillside behind the San Diego sewage-treatment plant (Nilsen et al., 1984). Most (50 m) of the reversed magnetozone, however, is located in a covered interval above the sea cliffs (which record normal polarity), and below the hillside section behind the San Diego sewage-treatment plant (Nilsen et al., 1984). Given the gentle northeast dip of the beds and the proximity to Sunset Cliffs, the reversed magnetozone at Sunset Cliffs corresponds to the reversed magnetozone partially covered at Point Loma. The reversed zone on Point Loma is therefore 30–50 m thick. The reversed sections at Point Loma and La Jolla are nearly equal in thickness, and both occur within similar mudstone facies on either side of the syncline (Fig. 3) and therefore are considered equivalent. This reconfirms previous lithostratigraphic (Nilsen and Abbott, 1981) and biostratigraphic (Sliter, 1968) correlations across the syncline.

AGE ASSIGNMENT

The Point Loma Formation reversed zone can be uniquely correlated to the magnetic polarity time scale on the basis of available biostratigraphic data. Sliter (1968) proposed three foraminiferal assemblage zones based on numerous planktonic and several specialized benthonic species from the Point Loma Formation. These zones have been variously correlated (Fig. 4) to the extensively used Goudkoff (1945) foraminiferal zones defined from deposits of the Great Valley, California. Ammonite zones and lineages of bivalves and gastropods provide correlations with other Pacific Slope deposits of Late Cretaceous age from Alaska to Baja California, including those of the Great Valley section.

The Pacific slope ammonite zonation of Jeletzky (*in* Muller and Jeletzky, 1970) places the Campanian/Maastrichtian boundary between the zone of *Metaplacenticeras pacificum* (Smith) and the early Maastrichtian zone of *Pachydiscus (Neodesmoceras) catarinae* (Anderson & Hanna). Yeo's (1982) collection of *Metaplacenticeras* sp., which we were unable to replicate, from the lower sandstone at La Jolla suggests that the lowermost exposed Point Loma Forma-

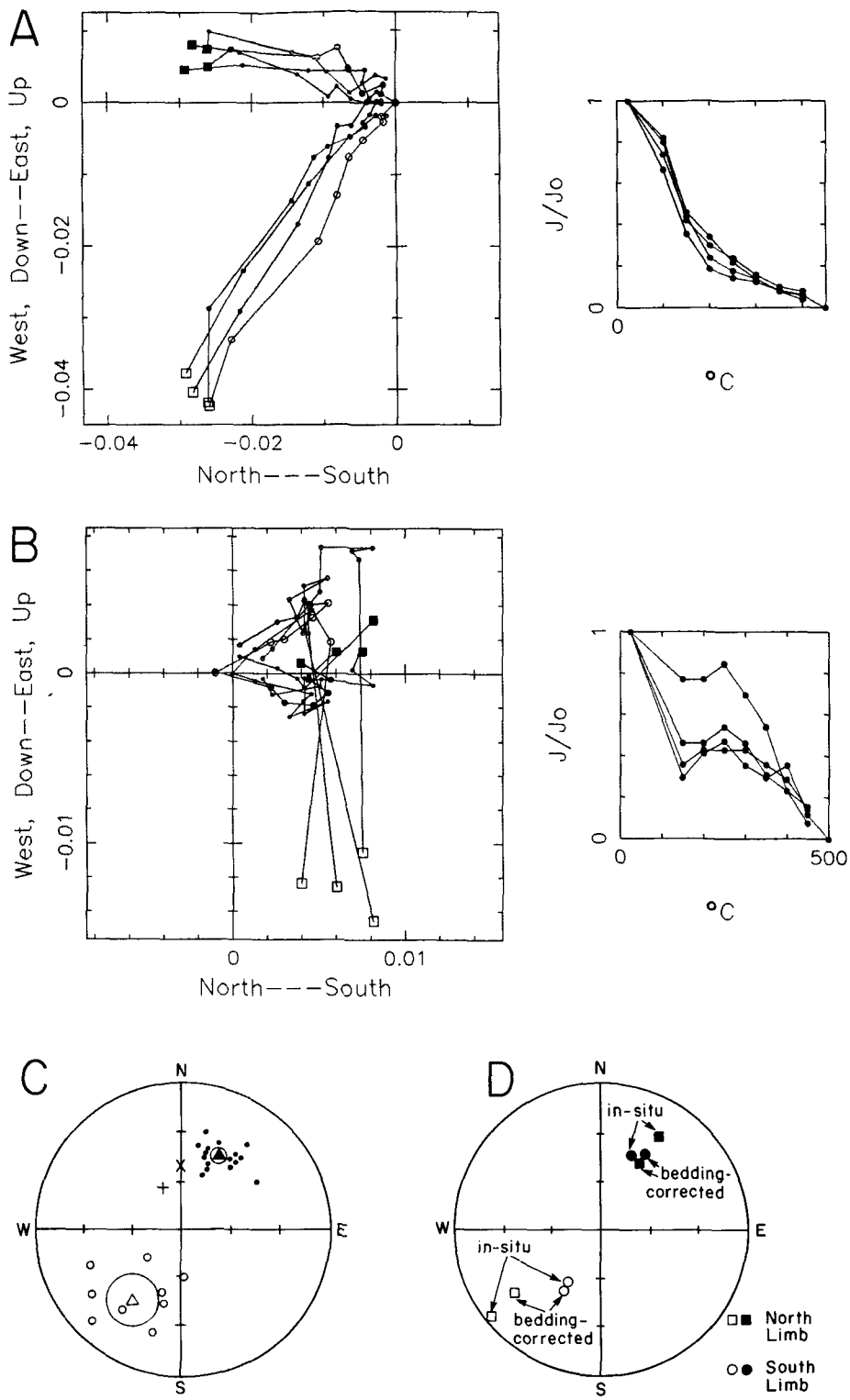


Figure 2. Modified Zijderveld demagnetization diagrams. Solid symbols plot declination vectors, open symbols plot inclination vectors. Axes rotated 270°. A: Class 1 normal. B: Class 1 reversed. C: Stereonet projection of class 1 bedding-corrected data. Normal inclination (I) = 39.5°, declination (D) = 28.0° (α_{95} = 5.4°). Reversed I = 36.4°, D = -145.1° (α_{95} = 16.6°). D: In situ and bedding-corrected averages separated north and south of the Mission Bay fold axis.

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tion is of late Campanian age. Mollusks from this lower sandstone (Fig. 4) include *Baculites* aff. *B. rex* Anderson, *Glycymeris* (*Glycymerita*) *anae* Smith, and *Calva peninsularis* (Anderson & Hanna). All occur elsewhere with the relatively low sculptured, late forms of *M. pacificum*. All except *G. (G.) anae* have been found also above *Metaplacenticeras* and associated with early Maastrichtian *P. (N.) catarinae*. These mollusks suggest a latest Campanian age for the basal sandstone beds. In contrast, mollusks from the overlying 6 m of mudstone at La Jolla (Fig. 4) include *Baculites occidentalis* Meek and *Inoceramus (Cataceramus)* aff. *I. (C.) goldfussianus* d'Orbigny, all of which suggest a late Campanian–early Maastrichtian age.

The earliest Maastrichtian ammonite indicators for the Nanaimo Basin, British Columbia and for California are *Nostoceras hornbyense* and *Pachydiscus* (*N.*) *catarinae* (Jeletzky, *in* Muller and Jeletzky, 1970). Megafossils from the Point Loma and overlying Cabrillo Formations on Point Loma (Fig. 4) supporting an early Maastrichtian age include *P.* (*N.*) *catarinae*, *Nostoceras* aff. *N. hornbyense* (Whiteaves), *Baculites lomaensis* Anderson, *Perissitys colocara* Popenoe & Saul, *Biplica miniplicata* Popenoe, *Crassastella lomana* (Cooper), and *Loxo decore* Dailey & Popenoe. *Nostoceras* aff. *N. hornbyense* and *B. lomaensis* from these strata occur within and above the magnetic reversal. *B. lomaensis* occurs with *P. (N.)* aff. *P. (N.) catarinae* above *Metaplacenticeras* aff. *M. pacificum* near the top of the Chatsworth Formation, Simi Hills, California (Saul and Alderson, 1981), from which only E zone foraminifera have been identified (Almgren, 1981). *B. lomaensis* also occurs with *P. (N.)* aff. *P. (N.) catarinae* in the "Ragged Valley Shale" along Los Gatos Creek, north of Coalinga, California, above *Metaplacenticeras* from the Alcalde Shale. The Ragged Valley Shale strata that yield *B. lomaensis* and other mollusks associated with it in the Point Loma Formation also yield Goudkoff E zone foraminifera (Almgren, 1986).

Whereas foraminiferal correlation of Sliter's (1968) Rosario Group zones with Goudkoff's Great Valley zones has been impeded by differences in paleoenvironmental conditions between the two areas (Sliter, 1979, 1986), and thus remains tentative (Sliter, 1984), the mollusks can be readily correlated. They support a correlation to the Lower E zone (Almgren, 1986) for the basal sandstone of the Point Loma Formation and to the E zone for much of the Point Loma Formation, including that portion with the magnetic reversal. Although Rosario Group strata correlative with Goudkoff's D_2 zone may occur in the San Diego area, mollusks restricted to that zone have not been recognized there. In

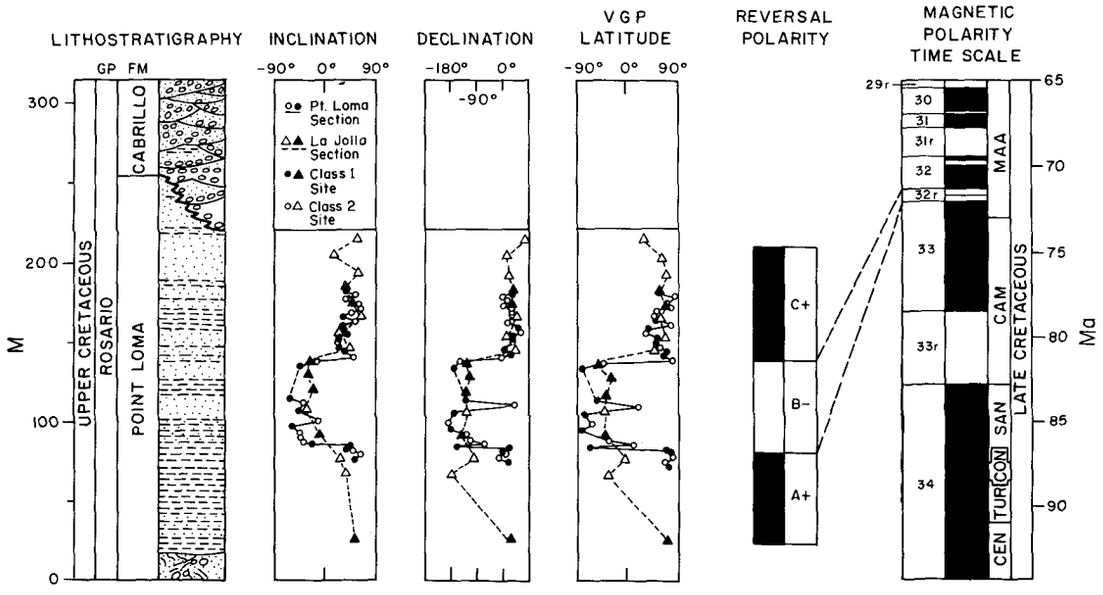


Figure 3. Summary diagram of San Diego paleomagnetic data showing correlation to magnetic polarity time scale (Harland et al., 1982). Solid lines connect Point Loma data, dashed lines connect La Jolla data. VGP = virtual geomagnetic pole.

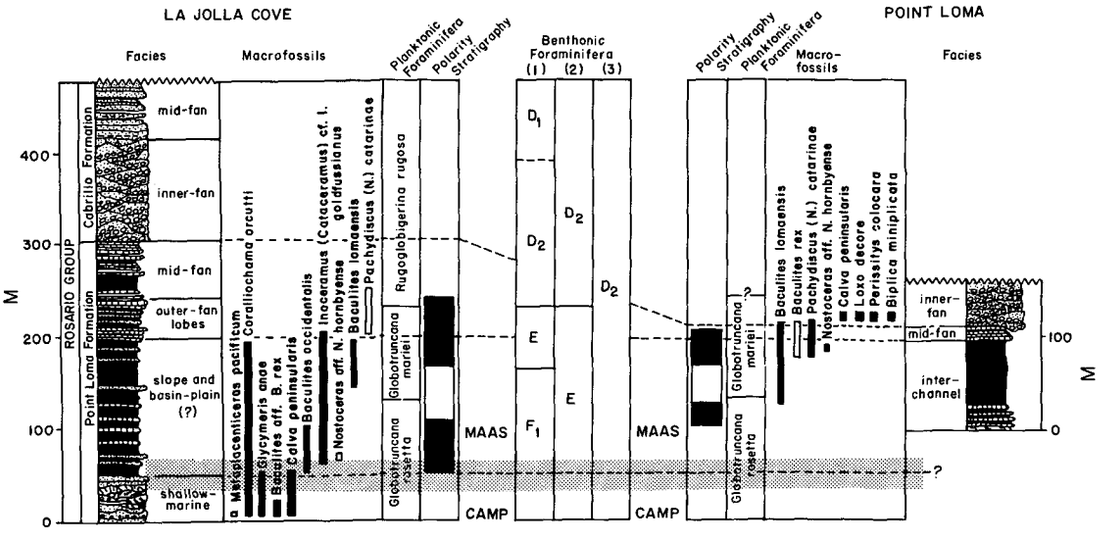


Figure 4. Lithostratigraphic, biostratigraphic, and magnetostratigraphic correlations of La Jolla Cove and Point Loma sections. Facies determination from Nilsen and Abbott (1981). Stippled zone marks interval in which Campanian/Maastrichtian boundary is located. Black bar ranges of selected macrofossils based on specimens in collections of Natural History Museum of Los Angeles County and San Diego Natural History Museum (includes specimens reported by Dawson [1978] and Sundberg [1984]); white bar ranges from occurrences reported by Yeo (1982) (specimens have not been available to us). Planktonic foraminifera after Sliter (1968). Benthonic foraminifera after (1) Sliter (1968), (2) Almgren (1973), and (3) Popenoe et al. (1960).

previous studies the Pacific slope Campanian/Maastrichtian boundary has, in general, been placed between Goudkoff's E and D_2 zones (e.g., Popenoe et al., 1960; Sliter, 1968; Almgren, 1973, 1981). If such a placement were correct, most of the Point Loma Formation would be assigned to the late Campanian. However, because all of the biostratigraphic interpretations limit the Point Loma strata to a mid-Campanian to early Maastrichtian age, we can uniquely correlate this reversed magnetozone to early Maastrichtian (e.g., Harland et al., 1982) Chron 32r. The reversed zone cannot be the early Campanian Chron 33r, which occurs within the zone of *Baculites chicoensis* (Ward et al., 1983). The Point Loma Formation reversal overlaps the stratigraphic occurrence of *Baculites lomaensis*, a species with a narrow range within the *Pachydiscus (N.) catarinae* zone and within Goudkoff's E zone. Although the E zone has been considered almost universally to be entirely late Campanian in age, the presence of the reversal concurrent with *B. lomaensis* places the upper part of the E zone in the Maastrichtian and also suggests that this reversal is early Maastrichtian Chron 32r.

CONCLUSIONS

Identification of reversed Chron 32r in the Point Loma Formation is particularly significant because the different placements of the macrofossil and microfossil Campanian/Maastrichtian stage boundaries have continually resulted in erroneous correlations of Pacific Slope strata. The magnetostratigraphic results support placement of the boundary between the zones of *Metaplacenticeras pacificum* and *Pachydiscus (Neodesmoceras) catarinae* and indicate the need for reassessment of the microfaunal zonation.

The improved age estimates for the Point Loma Formation require a revision of the overall depositional-tectonic history of Upper Cretaceous strata within the Peninsular Ranges terrane (Bottjer and Link, 1984; Morris et al., 1986). Previous studies of the differential contribution of eustatic and local tectonic processes in the development of Upper Cretaceous stratigraphic sequences in southern California (Yeo, 1982; Bottjer and Link, 1984) concluded that eustasy was the major control for these apparently similar sequences. Such an interpretation was supported for the Upper Cretaceous in part by the mid- to late Campanian microfossil age determinations reported for much of the lower Point Loma Formation (e.g., Sliter, 1979). These allowed apparent correlation with strata of megafossil-based Campanian age occurring in the next large, well-studied Upper Cretaceous outcrop region to the north (northern Santa Ana Mountains), some of which were deposited in slope environments (Buck and Bottjer, 1985). This correlation encouraged a scenario of regional eustatic control of depositional patterns. However, only Turonian-Campanian strata have been reported from the northern Santa Ana Mountains, although Maastrichtian strata might have existed there before the post-Cretaceous erosion event that preceded deposition of Cenozoic strata (Bottjer and Link, 1984). Thus, this new resolution of the position of the Campanian/Maastrichtian boundary in the San Diego Cretaceous indicates that, rather than a strong correlation in age and environment of deposition between these two major outcrop areas of the Peninsular Ranges terrane, there is actually little or perhaps no overlap in age. Therefore, although eustasy may have played a strong role in the development of stratigraphic sequences once deposition was initiated, local tectonics controlled the initiation and duration of sedimentation for this terrane. Continued magnetostratigraphic-biostratigraphic studies of west coast North American Upper Cretaceous strata are needed to further resolve the differing contribution of eustasy and regional tectonics toward development of the observed stratigraphic sequences.

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